

Extended Geometrical Perturbation Based Detectors for PolSAR Image Target Detection in Heterogeneously Patched Regions

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Abstract—Target detection in synthetic aperture radar image utilizing polarimetric information has attracted considerable attention. Single-target detector (STD), partial-target detector (PTD), and geometrical perturbation-polarimetric notch filter (GP-PNF) are three traditional polarimetric detectors based on polarimetric information. STD aims at detecting single targets, whereas PTD is suitable for partial targets. GP-PNF focuses on detecting targets with features, which are different from the homogeneous background. Both STD and PTD need a prior knowledge of the target, whereas GP-PNF needs to estimate the local clutter automatically. All these three methods use a feature vector to describe the character of the target or clutter. In fact, the feature vectors of the clutter and target may distribute in a subspace. Especially for the heterogeneous background, a feature vector cannot accurately describe the clutter. Motivated by this, this paper extends the clutter model from a complex feature vector to a complex feature subspace, which is suitable for a heterogeneously patched region and derives extended PTD and extended GP-PNF. Experimental results show the extended detectors' validation and superiority to traditional detectors for target detection in heterogeneous regions.

Index Terms—Heterogeneous region, radar polarimetry, synthetic aperture radar (SAR), target detection.

I. INTRODUCTION

THE polarimetric synthetic aperture radar (PolSAR), which can acquire the full polarization information and the imaging scene, becomes a mainstream in microwave remote sensing [1]. Many advanced polarimetric techniques have been proposed. Polarimetric target decomposition is a powerful technique to interpret scattering mechanisms in PolSAR data and is widely used in unsupervised target classification [2], [3]. It contains two main categories: eigenvalue eigenvector based decomposition [4], [5] and model-based decomposition [6], [7]. Chen *et al.* [8] developed the uniform polarimetric matrix rotation theory. Utilizing the theory, Xiao *et al.* [9] proposed polarimetric

coherence optimization for manmade target extraction. Based on the uniform polarimetric matrix rotation theory, Chen [10] also proposed a visualization and characterization tool called polarimetric coherence pattern.

This paper focuses on target detection exploiting polarimetric information. The polarization of the electromagnetic field scattered by an object keeps unique discrimination information about the scatterers, which can be exploited to detect specific targets [11]. Generally, target detection algorithms based on polarimetric information can be roughly divided into two categories [12], namely the statistics information based algorithms [13]–[15] and physical mechanism based algorithms [12], [16].

Statistics information based algorithms include optimal polarimetric detector (OPD) [13], [15] and identity likelihood-ratio test (ILR) [13], [14]. OPD requires a priori information about the mean and correlation matrix of the target and the clutter; these must be adjusted to the different scenarios before any detection. ILR is a variant of OPD, where the target correlation matrix is substituted with a scaled identity matrix and the target mean is substituted with zero. ILR requires a priori knowledge of the clutter correlation matrix and the ratio between the target and the clutter.

Physical mechanism based algorithms include single-channel detector, span detector, and power maximization synthesis (PMS) [12], [13]. The single-channel detector exploits power of one polarimetric channel, such as HH channel or HV channel. The benefit of the single-channel detector is the low complexity of the system, whereas the drawback is the performance is rather poor in terms of missed detections and false alarms. The span detector reduces the rate of missed detection due to unfortunate choice of the antenna polarization by considering the total power acquired in the scattering matrix. PMS is proposed as an improvement to the span detector. It is a function of the components of the measurement vector and makes no use of a priori target or clutter statistics.

Polarimetric whitening filter (PWF) is an algorithm that do not belong to neither of the two categories [12], [13]. It is based on statistical signal processing of quad polarimetric data but it does not employ any priori statistical information.

Single-target detector (STD), partial-target detector (PTD), and geometrical perturbation-polarimetric notch filter (GP-PNF) are three detector physical mechanism based algorithms. But different from aforementioned physical mechanism based algorithms, they are based on geometrical perturbation. STD

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mainly focuses on the detection of single targets that can be completely characterized by one Sinclair matrix, which can be modified as a three-dimensional (3-D) vector in the monostatic case [17]–[19]. The vector is usually normalized and stands for the scattering mechanism of the target. Hence, the vector only represents the relative information between different scattering values, which is independent of the scattering intensity. In 2010, Marino *et al.* extended STD to PTD which aims at detecting partial targets [11], [18], [20]. Similar to STD, PTD describes a target with a correlation matrix which can be modified as a 6-D vector in the monostatic case [11], [18], [20]. The vector is also normalized and stands for the scattering mechanism of a partial target. Both STD and PTD are constructed based on the sensitivity of the polarimetric complex coherence to changes in polarization, and both of them need a prior knowledge of the detected targets. In 2010, Marino proposed GP-PNF [21]–[28]. GP-PNF rejects the selected partial target and detects anything different from it. Apparently, GP-PNF needs to locally estimate the background clutter using large averaging windows. It has been successfully applied in ship detection in the sea clutter. However, it is suitable only for locally homogeneous background. GP-PNF will fail when detecting targets in a heterogeneous background. That is because the heterogeneous region cannot be accurately described by a correlation matrix or a 6-D vector since the scattering mechanisms in the region are very different. To clarify, with heterogeneous clutter background we mean a background that is extended on large image regions, and it is composed by several patches. Each patch is characterized by a different partial target.

To overcome the deficiency of GP-PNF, extended PTD (EPTD) and extended GP-PNF (EGP-PNF) are proposed in this paper. Different from GP-PNF, EPTD and EGP-PNF can detect targets in heterogeneously patched regions. These two detectors use a subspace to represent the clutter. The subspace is the lowest dimensional subspace that contains the vectors, which stands for the scattering mechanism in the region. Then, we construct EPTD and EGP-PNF, which allow the detection of targets in a heterogeneous region based on the sensitivity of the polarimetric complex coherence. Experimental results show that EPTD and EGP-PNF outperform PTD, GP-PNF, and PWF on detecting targets in a heterogeneous background.

This paper is structured as follows. STD, PTD, and GP-PNF are briefly reviewed in Section II. The proposed EPTD and EGP-PNF are derived in Section III. The analysis of the proposed two detectors is presented in Section IV. The experiment results and the conclusion are presented in Sections V and VI, respectively.

II. TRADITIONAL DETECTORS

A. Single-Target Detector

A single target is any target scattering an electromagnetic wave with a fixed polarization in time or space [1]. It can be characterized using the scattering matrix

$$[S] = \begin{bmatrix} HH & HV \\ VH & VV \end{bmatrix} \quad (1)$$

where H stands for linear horizontal and V for linear vertical. The Sinclair matrix can be represented by the vector built as follows [1]:

$$\underline{k} = \frac{1}{2} \text{Trace}([S] \Psi) = [k_1, k_2, k_3, k_4]^T \quad (2)$$

where $\text{Trace}(\cdot)$ is the trace of the matrix, and Ψ is a complete set of 2×2 basis matrices under a Hermitian inner product. In the monostatic case, the scattering matrix is symmetric (i.e., $HV = VH$) and only three complex numbers are needed to characterize a single target [1]. Therefore, 4-D scattering vector can be substituted with a 3-D one. For the sake of brevity, in this paper, the hypothesis of a symmetric scattering matrix will be assumed.

A scattering mechanism can be defined as a normalized vector $\underline{\omega} = \underline{k}/\|\underline{k}\|$ [12], [17]–[19]. When a target (i.e., scattering mechanism) is selected, its backscattering can be determined as [12], [17]–[19]

$$i(\underline{\omega}) = \underline{\omega}^H \underline{k}. \quad (3)$$

When two scattering mechanisms $\underline{\omega}_1$ and $\underline{\omega}_2$ are selected, two different images can be extracted from the observables. Then, the polarimetric coherence is defined as [12], [17]–[19]

$$\gamma = \frac{\langle i(\underline{\omega}_1) i(\underline{\omega}_2)^* \rangle}{\sqrt{\langle i(\underline{\omega}_1) i(\underline{\omega}_1)^* \rangle \langle i(\underline{\omega}_2) i(\underline{\omega}_2)^* \rangle}} \quad (4)$$

where $\langle \cdot \rangle$ is the finite averaging operator. The operator is a box-car filter that averages the pixels in a local window [29], [30]. Note that this coherence is only polarimetric and not interferometric [19]. That is, we are not employing interferometry but rather a single flight pass [19].

Then, an STD can be constructed [12], [17]–[19]. First, a basis for target space is defined so that the target of interest lies exclusively on one component of the 3-D complex vector $\underline{k}$. In the following, the scattering mechanism after the change of basis is regarded as $\underline{\omega}_T = [1, 0, 0]^T$. The resulting image when the target $\underline{\omega}_T$ is selected is

$$i(\underline{\omega}_T) = k_1. \quad (5)$$

In (5), the second and the third components of the scattering vector disappear, since the target of interest lies exclusively in k_1 . For this reason, k_2 and k_3 are considered as clutter. The second scattering mechanism $\underline{\omega}_P$ (i.e., perturbed target) is obtained by rotating slightly the vector $\underline{\omega}_T$ in the polarimetric space. The same change of basis that makes $\underline{\omega}_T = [1, 0, 0]^T$ is performed on $\underline{\omega}_P$. Consequently, $\underline{\omega}_P = [a, b, c]^T$ is obtained, with a, b , and c being the complex numbers. Considering $\underline{\omega}_T \approx \underline{\omega}_P$, we have $|a| \approx 1$, $|b| \approx 0$, and $|c| \approx 0$. Then, the polarimetric coherence can be estimated as [12], [17]–[19]

$$\gamma(\underline{\omega}_T, \underline{\omega}_P) = \frac{\langle i(\underline{\omega}_T) i(\underline{\omega}_P)^* \rangle}{\sqrt{\langle i(\underline{\omega}_T) i(\underline{\omega}_T)^* \rangle \langle i(\underline{\omega}_P) i(\underline{\omega}_P)^* \rangle}} \quad (6)$$

where

$$\begin{aligned}
 \langle i(\underline{\omega}_T) i^*(\underline{\omega}_P) \rangle &= a \langle |k_1|^2 \rangle + b \langle k_1 k_2^* \rangle + c \langle k_1 k_3^* \rangle \\
 \langle i(\underline{\omega}_T) i^*(\underline{\omega}_T) \rangle &= \langle |k_1|^2 \rangle \\
 \langle i(\underline{\omega}_P) i^*(\underline{\omega}_P) \rangle &= |a|^2 \langle |k_1|^2 \rangle + |b|^2 \langle |k_2|^2 \rangle + |c|^2 \langle |k_3|^2 \rangle \\
 &\quad + 2\text{Re}(ab^* \langle k_1 k_2^* \rangle) + \text{Re}(ac^* \langle k_1 k_3^* \rangle) \\
 &\quad + 2\text{Re}(cb^* \langle k_3 k_2^* \rangle). \tag{7}
 \end{aligned}$$

It should be noticed that $|\gamma|$ is large when $\underline{k}$ is close to the target scattering mechanism and small when $\underline{k}$ is very different from the target scattering mechanism. This holds only when the perturbed target $\underline{\omega}_P$ is very close to $\underline{\omega}_T$. This is why the perturbation is assumed to be small.

However, the cross products between the components of the scattering vector introduce biases, in case the components are correlated with each other [12], [19]. So the cross products must be neglected [12], [19]. Then, the detector is obtained [12], [17]–[19]

$$\begin{aligned}
 \gamma_d &= |\gamma(\underline{\omega}_T, \underline{\omega}_P)| \\
 &= \frac{|a| \langle |k_1|^2 \rangle}{\sqrt{\langle |k_1|^2 \rangle (|a|^2 \langle |k_1|^2 \rangle + |b|^2 \langle |k_2|^2 \rangle + |c|^2 \langle |k_3|^2 \rangle)}}. \tag{8}
 \end{aligned}$$

Dividing both numerator and denominator by $|a| \langle |k_1|^2 \rangle$ and defining the power as $P_T = \langle |k_1|^2 \rangle$, $P_{C2} = \langle |k_2|^2 \rangle$, and $P_{C3} = \langle |k_3|^2 \rangle$, the expression of the detector can be simplified as follows:

$$\gamma_d = \frac{1}{\sqrt{1 + \frac{|b|^2}{|a|^2} \frac{P_{C2}}{P_T} + \frac{|c|^2}{|a|^2} \frac{P_{C3}}{P_T}}}. \tag{9}$$

Here, $(|b|/|a|)^2$ and $(|c|/|a|)^2$ are regarded as reduction ratios (RedRs).

The optimal choice of the perturbed target in the absence of a priori information about clutter is $|b| = |c|$ [12], [19]. If the total power of the clutter is indicated with $P_C = P_{C2} + P_{C3}$, the detector is further simplified as follows:

$$\gamma_d = \frac{1}{\sqrt{1 + \text{RedR}_{\frac{P_C}{P_T}}}}. \tag{10}$$

The final expression of the detector sets a threshold on (10)

$$\gamma_d(\underline{\omega}_T, \underline{\omega}_P) \geq T. \tag{11}$$

B. Partial-Target Detector

PTD is an extension of STD [11], [12], [18], [20]. First, a partial scattering vector is defined as follows: [11], [12]

[18], [20]

$$\begin{aligned}
 \underline{t} &= \text{Trace}([C] \Psi) \\
 &= [t_1, t_2, t_3, t_4, t_5, t_6]^T \\
 &= [\langle |k_1|^2 \rangle, \langle |k_2|^2 \rangle, \langle |k_3|^2 \rangle, \langle k_1^* k_2 \rangle, \langle k_1^* k_3 \rangle, \langle k_2^* k_3 \rangle]^T \tag{12}
 \end{aligned}$$

where Ψ is a set of 6×6 basis matrices under a Hermitian inner product. The $\underline{t}$ vector belongs to a subspace of $\mathbb{C}^6$. In particular, the partial target to be detected and the perturbed target are regarded as follows:

$$\begin{aligned}
 \hat{\underline{t}}_T &= \text{Trace}([C_T] \Psi) / \|\text{Trace}([C_T] \Psi)\| \\
 \hat{\underline{t}}_P &= \text{Trace}([C_P] \Psi) / \|\text{Trace}([C_P] \Psi)\|. \tag{13}
 \end{aligned}$$

A change of basis is performed which makes the target of interest lie only in one nonzero component:

$$\hat{\underline{t}}_T = [1, 0, 0, 0, 0, 0]^T \quad \hat{\underline{t}}_P = [a, b, c, d, e, f]^T. \tag{14}$$

Suppose the elements on the diagonal of $[A]$ are the components of the partial scattering vector $\underline{t}$ after the change of basis which makes $\hat{\underline{t}}_T = [1, 0, 0, 0, 0, 0]^T$. The detector can be achieved with [11], [12], [18], [20]

$$([A] \hat{\underline{t}}_T)^H ([A] \hat{\underline{t}}_P) = \hat{\underline{t}}_T^H [A]^H [A] \hat{\underline{t}}_P = \hat{\underline{t}}_T^H [P] \hat{\underline{t}}_P \tag{15}$$

where

$$[P] = [A]^H [A] = \text{diag}(P_1, P_2, P_3, P_4, P_5, P_6). \tag{16}$$

Then, the detector is obtained [11], [12], [18], [20]:

$$\begin{aligned}
 \gamma_d(\underline{t}_T, \underline{t}_P) &= \frac{\hat{\underline{t}}_T^H [P] \hat{\underline{t}}_P}{\sqrt{(\hat{\underline{t}}_T^H [P] \hat{\underline{t}}_T) (\hat{\underline{t}}_P^H [P] \hat{\underline{t}}_P)}} \\
 &= \frac{1}{\sqrt{1 + \frac{b^2}{a^2} \frac{P_2}{P_1} + \frac{c^2}{a^2} \frac{P_3}{P_1} + \frac{|d|^2}{a^2} \frac{P_4}{P_1} + \frac{|e|^2}{a^2} \frac{P_5}{P_1} + \frac{|f|^2}{a^2} \frac{P_6}{P_1}}}. \tag{17}
 \end{aligned}$$

Specifically, in the absence of a priori information about the clutter, the perturbed target is chosen as [11], [12], [18], [20]

$$b = c = |d| = |e| = |f|. \tag{18}$$

If we define the clutter power as $P_C = P_2 + P_3 + P_4 + P_5 + P_6$, the target power as $P_T = P_1$, and the RedR as $\text{RedR} = (b/a)^2$, the detector becomes

$$\gamma_d = \frac{1}{\sqrt{1 + \text{RedR}_{\frac{P_C}{P_T}}}}. \tag{19}$$

C. Geometrical Perturbation-Polarimetric Notch Filter

GP-PNF is proposed by Marino for ship detection [21]–[28]. Because of its local homogeneity, the sea clutter can be characterized by a partial scattering vector. PTD is able to detect the homogeneous clutter. Then, one can extract the targets that are not the homogeneous clutter. Based on this idea, Marino develops GP-PNF. The idea behind the GP-PNF is to reject the sea return and extract the remaining features [23].

The mathematical derivation details of the GP-PNF can be found in [23] and [26]. Here, only the final detector expression is presented for the sake of brevity.

$$\gamma_d = \frac{1}{\sqrt{1 + \frac{\text{RedR}}{\underline{\hat{t}}^{*T} \underline{\hat{t}} - |\underline{\hat{t}}^{*T} \underline{\hat{t}}_{sea}|^2}}} \quad (20)$$

where $\underline{\hat{t}}_{sea}$ is the normalized partial scattering vector of the sea clutter, and RedR is the reduction ratio.

III. PROPOSED EXTENDED GEOMETRICAL PERTURBATION BASED DETECTORS

A. Construction of EPTD

GP-PNF describes the locally homogeneous clutter with a partial scattering vector. However, only one partial scattering vector is not sufficient to describe a heterogeneous region. Because a heterogeneous region is a region composed by different image patches, each of which characterized by a different partial target. On the other hand, the heterogeneous clutter can always be characterized by a subspace with a sufficient large dimension. This fact can be demonstrated as follows. Let $\underline{\hat{t}}_C$ be the normalized partial scattering vector of the heterogeneous clutter. Since each patch of the heterogeneous clutter has a different partial scattering vector, $\underline{\hat{t}}_C$ is treated as a random vector. We use the principal components analysis method to analyze $\underline{\hat{t}}_C$. Let C be the correlation matrix of $\underline{\hat{t}}_C$

$$[C] = E[\underline{\hat{t}}_C^H \underline{\hat{t}}_C] \quad (21)$$

where $E[\cdot]$ means expectation. Let $v_1, v_2, \dots, v_m$ be the eigenvectors corresponding to the first m largest eigenvalues. And assume that the remaining eigenvalues are small enough compared with the m largest eigenvalues, so that they can be neglected. Then, subspace $\text{span}\{v_1, v_2, \dots, v_m\}$ can be used to represent the heterogeneous clutter, because all the partial scattering vectors of the heterogeneous clutter lie in the subspace. Apparently, when the dimension of the subspace is 1, the clutter model is equivalent to the case in PTD.

Now EPTD can be constructed. First, select a target of interest represented by a subspace S_T and a perturbed target represented by S_P . Both S_T and S_P are m dimensional. Similar to the case presented in Section II-A, we assume that the perturbation is small. Hence the distance between S_T and S_P is small. The distance between two subspaces can be measured in terms of their principal angles. For m -dimensional subspaces S_T and S_P , there exist m principal angles $\{\theta_i\}_{i=1}^m$ and associated principal vectors $\{\underline{u}_i, \underline{v}_i\}_{i=1}^m$ defined as [31]

$$\cos(\theta_k) = \underline{u}_k^T \underline{v}_k = \max_{\substack{\underline{u} \in S_T, \|\underline{u}\|_2=1 \\ \underline{u}^T [\underline{v}_1, \dots, \underline{v}_{k-1}] = 0}} \max_{\substack{\underline{v} \in S_P, \|\underline{v}\|_2=1 \\ \underline{v}^T [\underline{u}_1, \dots, \underline{u}_{k-1}] = 0}} \underline{u}^T \underline{v}. \quad (22)$$

Then, the distance between the two subspaces can be defined as follows [31]:

$$d(S_T, S_P) = \left[\sum_{i=1}^m \sin^2(\theta_i) \right]^{\frac{1}{2}}. \quad (23)$$

It is noted that $d(S_T, S_P)$ is small when θ_i is small and vice versa.

Let $[W_T]$ and $[W_P]$ be the standard orthogonal basis matrix of the subspaces of S_T and S_P , respectively. And suppose the columns of $[W_T]$ and $[W_P]$ are the principal vectors of S_T and S_P , respectively

$$\begin{aligned} [W_T] &= [u_1, u_2, \dots, u_m] \\ [W_P] &= [v_1, v_2, \dots, v_m]. \end{aligned} \quad (24)$$

The principal vectors between two subspaces can be computed using the QR factorization and the singular value decomposition (SVD) [31].

Once the target matrix $[W]$ is determined, the backscattering can be determined as follows:

$$\underline{i}([W]) = [W]^H \underline{t} \quad (25)$$

where $\underline{t}$ is the partial scattering vector. It should be noted that (3) is a special case of (25). $\underline{i}([W])$ is an m -dimensional vector, whereas $i(\underline{\omega})$ is a complex value. Then, the generalized polarimetric coherence can be defined as follows:

$$\gamma([W_T], [W_P]) = \frac{\underline{i}^H([W_T]) \underline{i}([W_P])}{\sqrt{\underline{i}^H([W_T]) \underline{i}([W_T])} \sqrt{\underline{i}^H([W_P]) \underline{i}([W_P])}} \quad (26)$$

It should be noted that $|\gamma|$ is large when $\text{range}(\underline{t})$ is close to the target scattering mechanism determined by the subspace S_T , whereas $|\gamma|$ is small when $\text{range}(\underline{t})$ is far from the subspace S_T . This property holds only when the perturbation is small. And this is why we assume that the perturbation is small.

Similar to the STD and PTD case, a change of basis is performed on $[W_T]$ and $[W_P]$. Let $[W_T, W_T^C]$ be the changed standard orthogonal basis matrix of polarimetry space, where $[W_T^C]$ is the standard orthogonal basis matrix of the orthogonal complement subspace of S_T . $[W_T^C]$ can be built according to the following procedure. Let

$$P_{S_T^\perp} = I - [W_T][W_T]^H \quad (27)$$

where $P_{S_T^\perp}$ is the orthogonal projection matrix (OPM) of the orthogonal complement subspace of S_T . Perform eigenvalue decomposition on $P_{S_T^\perp}$. The columns of W_T^C are exactly the eigenvectors corresponding to the eigenvalue 1.

After the change of basis, the standard orthogonal bases of S_T and S_P become

$$\begin{aligned} [W'_T] &= [W_T, W_T^C]^{-1} [W_T] = [u'_1, u'_2, \dots, u'_m] \\ [W'_T]_{ij} &= \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases} \\ [W'_P] &= [W_T, W_T^C]^{-1} [W_P] = [v'_1, v'_2, \dots, v'_m]. \end{aligned} \quad (28)$$

And the new representation of the partial scattering vector is

$$\underline{t}^{\text{trans}} = [W_T, W_T^C]^{-1} \underline{t}. \quad (29)$$

Suppose $\underline{t}^{\text{trans}} = [t_1, t_2, \dots, t_n]^T$. Since θ_i is very small, then

$$\cos(\theta_i) = u_i'^H v_i' \approx 1 \quad (30)$$

and

$$|v'_{ii}| \gg |v'_{ij}| \quad i \neq j. \quad (31)$$

The detector is constructed as

$$\begin{aligned} \gamma_d &= |\gamma([W_T], [W_P])| \\ &= |\gamma([W'_T], [W'_P])| \\ &= \frac{\left| \sum_{i=1}^m t_i^* \sum_{j=1}^n v_{ij} t_j \right|}{\sqrt{\sum_{i=1}^m |t_i|^2} \sqrt{\sum_{i=1}^m \sum_{j=1}^n \sum_{l=1}^n v'_{ij} v'^*_{il} t_j t_l^*}} \end{aligned} \quad (32)$$

where n is the dimension of $\underline{t}^{\text{trans}}$.

As mentioned in Section II-A, the cross products between the components of the scattering vector introduce biases [12], [19]. Hence, the cross products must be neglected. Then

$$\begin{aligned} \left| \sum_{i=1}^m t_i^* \sum_{j=1}^n v'_{ij} t_j \right| &\approx \sum_{i=1}^m |v'_{ii}| |t_i|^2 \\ \sqrt{\sum_{i=1}^m \sum_{j=1}^n \sum_{l=1}^n v'_{ij} v'^*_{il} t_j t_l^*} &\approx \sqrt{\sum_{j=1}^n \left(\sum_{i=1}^m |v'_{ij}|^2 \right) |t_j|^2}. \end{aligned} \quad (33)$$

And the detector becomes

$$\gamma_d = \frac{\sum_{i=1}^m |v_{ii}| |t_i|^2}{\sqrt{\sum_{i=1}^m |t_i|^2} \sqrt{\sum_{j=1}^n \left(\sum_{i=1}^m |v_{ji}|^2 \right) |t_j|^2}}. \quad (34)$$

Due to (31), there is

$$\sum_{i=1}^m |v_{ji}|^2 \approx |v_{jj}|^2 \approx 1 \quad j = 1, \dots, m. \quad (35)$$

To simplify the detector, set

$$\begin{aligned} |v_{jj}| &= a \quad j = 1, \dots, m \\ \sum_{i=1}^m |v_{ji}|^2 &= b_j^2 \quad j = m+1, \dots, n \end{aligned} \quad (36)$$

where a and b_j are positive real numbers. Then, the detector becomes

$$\gamma_d = \frac{1}{\sqrt{1 + \frac{\sum_{i=m+1}^n b_i^2 |t_i|^2}{a^2 \sum_{i=1}^m |t_i|^2}}}. \quad (37)$$

Specifically, in the absence of a priori information about the clutter, the perturbed target is chosen as

$$b_{m+1} = b_{m+2} = \dots = b_n = b. \quad (38)$$

Then, the final detector is obtained

$$\gamma_d = \frac{1}{\sqrt{1 + \text{RedR} \frac{\sum_{i=m+1}^n |t_i|^2}{\sum_{i=1}^m |t_i|^2}}} \quad (39)$$

where $\text{RedR} = b^2/a^2$. Equation (39) is exactly the extended version of PTD. If the total power of the target is defined as

$P_T = \sum_{i=1}^m |t_i|^2$ and the total power of the clutter is defined as $P_C = \sum_{i=m+1}^n |t_i|^2$, the detector can be expressed as follows:

$$\gamma_d = \frac{1}{\sqrt{1 + \text{RedR} \frac{P_C}{P_T}}}. \quad (40)$$

Equation (40) is formally the same as (10) and (19). Geometrically, P_T is the power that vector $\underline{t}$ lies in the target subspace; P_C is the power that $\underline{t}$ lies in the clutter subspace. It is noted that the clutter subspace is 1-D for PTD, which means the local homogeneity is needed. But EPTD described in (39) enables us to describe a more complicated region, such as a situation with two or more kinds of targets mixed together (e.g., forest and ground boundaries).

In fact, it is not necessary to change the basis when computing γ_d , as we did in (28) and (29). Denote the orthogonal projection matrices of subspaces S_T and S_C as $[Q_T]$ and $[Q_C]$, respectively. Then, (40) can be rewritten as

$$\gamma_d = \frac{1}{\sqrt{1 + \text{RedR} \frac{\|[Q_C]\underline{t}\|^2}{\|[Q_T]\underline{t}\|^2}}}. \quad (41)$$

where $\|\cdot\|$ stands for the 2-norm of a vector. It should be noted that (41) is independent of the choice of basis. If we perform a change of basis on Q_C , Q_T , and $\underline{t}$; suppose the change-of-basis matrix is $[U]$, which is a unitary matrix. Then, there is

$$\frac{1}{\sqrt{1 + \text{RedR} \frac{\|[U]^H [Q_C] U U^H \underline{t}\|^2}{\|[U]^H [Q_T] U U^H \underline{t}\|^2}}} = \frac{1}{\sqrt{1 + \text{RedR} \frac{\|[Q_C]\underline{t}\|^2}{\|[Q_T]\underline{t}\|^2}}}. \quad (42)$$

Hence, it is not necessary to change the basis when computing γ_d . All we need is to find the orthogonal projection matrices of subspaces S_T and S_C whatever the basis we choose.

Finally, a threshold T is set on (41)

$$\gamma_d \geq T. \quad (43)$$

EPTD is a stochastic process. To optimally set the threshold and assess the probability of false alarms and misdetection, the probability density function of the detector is required [19]. This is related to the statistical property of the clutter. However, Marino provides an expression of the detector independent of the statistical realization. [11], [19]. And this expression still applies to EPTD. Let $SCR = P_T/P_C$, where SCR is the signal-to-clutter ratio. Then, the detector becomes

$$\gamma_d = \frac{1}{\sqrt{1 + \text{RedR} \frac{1}{SCR}}} \geq T. \quad (44)$$

And it is equivalent to

$$SCR \geq \text{RedR} \frac{T^2}{1 - T^2}. \quad (45)$$

Proper RedR and T are chosen to detect targets with SCR larger than $\text{RedR} \frac{T^2}{1 - T^2}$.

B. Construction of EGP-PNF

As what is discussed in Section III-A, EPTD has a decision rule based on an SCR between target subspace and clutter

subspace. However, the balance between clutter and target defined as SCR can vary across the same scene [21]. In order to correct for these effects, the clutter backscattering has to be neglected in EPTD. Then, we obtain EGP-PNF

$$\gamma_d = \frac{1}{\sqrt{1 + \frac{\text{RedR}}{\| [Q_T] \underline{t} \|^2}}} = \frac{1}{\sqrt{1 + \frac{\text{RedR}}{\underline{t}^{*T} * \underline{t} - \| [Q_C] \underline{t} \|^2}}}. \quad (46)$$

Comparing (46) with (20), it is easy to be observed that GP-PNF is just the special case of EGP-PNF when the clutter subspace is 1-D. EGP-PNF removes the power in the clutter subspace and uses the power in the target subspace as the feature for target detection.

A threshold T is set on (46), and we have

$$\gamma_d = \frac{1}{\sqrt{1 + \frac{\text{RedR}}{\| [Q_T] \underline{t} \|^2}}} > T. \quad (47)$$

The detector is a stochastic process. To optimally set the threshold and assess the probability of false alarms and mis-detection, the probability density function of the detector is required, which is out of the scope of this paper. But we can still present an expression independent of the statistical realization. Equation (47) is equivalent to

$$\| [Q_T] \underline{t} \|^2 \geq \text{RedR} \frac{T^2}{1 - T^2}. \quad (48)$$

Proper RedR and T are chosen to detect targets with power larger than $\text{RedR} \frac{T^2}{1 - T^2}$ in the target space.

Now we discuss the relation between EGP-PNF and EPTD. If the partial scattering vector of the object $\underline{t}$ is normalized previously. Then (46) becomes

$$\begin{aligned} \gamma_d &= \frac{1}{\sqrt{1 + \frac{\text{RedR}}{\| [Q_T] \underline{t} \|^2}}} \\ &= \frac{1}{\sqrt{(1 + \text{RedR}) + \text{RedR} \frac{\| [Q_C] \underline{t} \|^2}{\| [Q_T] \underline{t} \|^2}}} \\ &= \frac{1}{\sqrt{(1 + \text{RedR})}} \cdot \frac{1}{\sqrt{1 + \frac{\text{RedR}}{(1 + \text{RedR})} \frac{\| [Q_C] \underline{t} \|^2}{\| [Q_T] \underline{t} \|^2}}} \end{aligned} \quad (49)$$

where $\hat{t} = \underline{t} / \|\underline{t}\|$. Obviously, EGP-PNF is equivalent to EPTD when $\underline{t}$ is normalized. The only difference between them is two constant coefficients. In other words, EPTD neglects the backscattering intensity and only cares about the scattering mechanism, whereas EGP-PNF cares about both the backscattering intensity and the scattering mechanism.

C. Determination of Target Subspace and Clutter Subspace

Determining the target subspace and the clutter subspace is equivalent to determining the relevant orthogonal projection matrices $[Q_T]$ and $[Q_C]$. In order to determine $[Q_T]$ and $[Q_C]$, training data are needed. Suppose there is a target dataset $\{\underline{t}_i\}_{i=1}^K$.

First, the correlation matrix of target should be estimated. The correlation matrix is estimated as follows:

$$[C_T] = \sum_{i=1}^K \hat{\underline{t}}_i \hat{\underline{t}}_i^H. \quad (50)$$

After the correlation matrix is obtained, eigenvalue decomposition is performed on $[C_T]$ and then

$$[C_T] = \sum_{i=1}^n \lambda_i \underline{v}_i \underline{v}_i^H \quad (51)$$

where $\lambda_i, i = 1, 2, \dots, n$, are the eigenvalues of $[C_T]$ satisfying $\lambda_1 \geq \dots \geq \lambda_n$, and $\underline{v}_i, i = 1, 2, \dots, n$, are the relevant eigenvectors. We set a threshold η satisfying $0 < \eta < 1$ and determine m subjecting to

$$\sum_{i=1}^m \lambda_i \geq \eta, \quad \sum_{i=1}^{m-1} \lambda_i < \eta. \quad (52)$$

Then, the target subspace S_T is $\text{span}(\underline{v}_1, \underline{v}_2, \dots, \underline{v}_m)$. And the OPM of S_T is

$$[Q_T] = [\underline{v}_1, \underline{v}_2, \dots, \underline{v}_m] \cdot [\underline{v}_1, \underline{v}_2, \dots, \underline{v}_m]^H. \quad (53)$$

Since S_C is the orthogonal complement subspace, the OPM of S_C can be determined as follows:

$$[Q_C] = I - [Q_T]. \quad (54)$$

Apparently, $[Q_C]$ can be first determined by using clutter training data and following the same procedures, thus $[Q_T]$ is determined by $[Q_T] = I - [Q_C]$.

Although the partial scattering vectors span a 6-D space, there only exist three orthogonal partial scattering vectors that are physically feasible. Hence, the dimension of the clutter subspace usually set to be less than 3. When the dimension of the clutter subspace is set to be larger than or equal to 3, the probability of detection may be reduced, since some target partial scattering vectors may be included in the clutter subspace.

D. Algorithm Implementation

This section demonstrates the detailed procedures of algorithm to detect targets in heterogeneous regions based on EPTD or EGP-PNF.

First, the partial target vector for each pixel of region of interest (ROI) is calculated.

Second, the sliding window method is exploited to detect the targets in ROI. The sliding window is shown in Fig. 1 [32]. Note that the choice of this window size here is for illustration purposes only. The cell marked in black is the pixels under test (PUTs). The boundary ring is shown in green, the cells in which are used to train the detector. The guard ring is shown in yellow. The target window size (PUTs) should be about the size of the smallest object that one wishes to detect; the guard ring window size should be about the size of the largest object; and the boundary ring window size should be large enough to estimate the subspace of the clutter accurately [32].

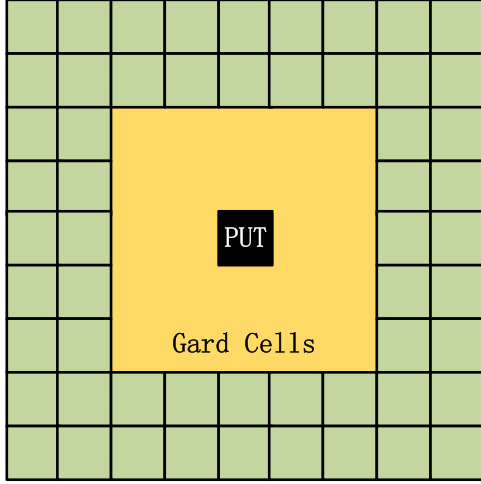


Fig. 1. Sliding window.

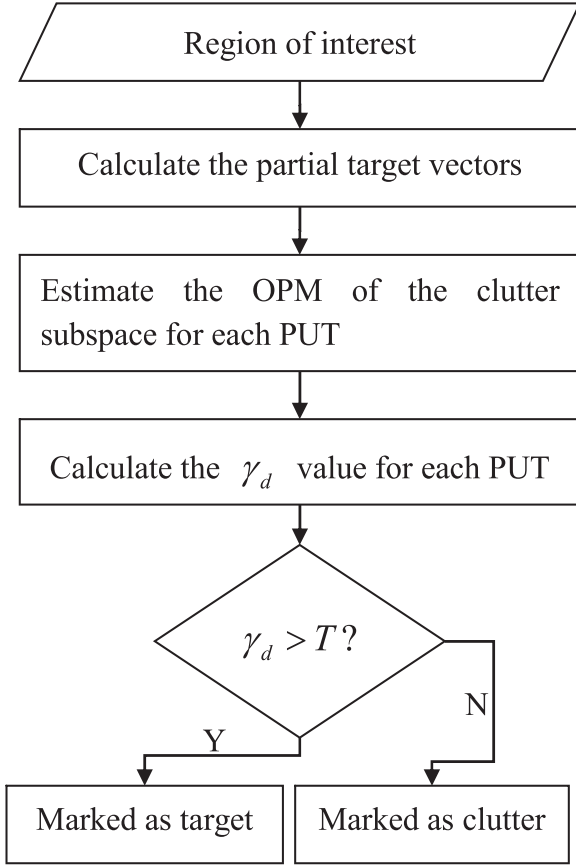


Fig. 2. Flowchart of algorithm to detect targets in heterogeneous regions based on EPTD or EGP-PNF.

When the window dimension is chosen, the window slides across the ROI. For PUTs, the OPM of clutter subspace is estimated using the pixels in the boundary ring by the method demonstrated in Section II-B. Then, the γ_d value for each PUT is calculated by (41). The PUTs that has γ_d values larger than the threshold T is target; whereas the PUTs that has γ_d values less than or equal to the threshold T is clutter.

The flowchart of this algorithm is summarized in Fig. 2.

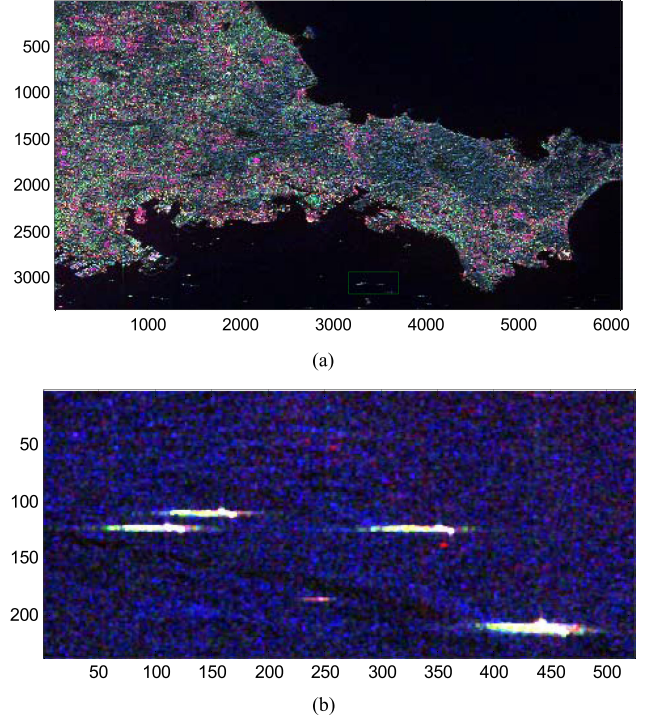
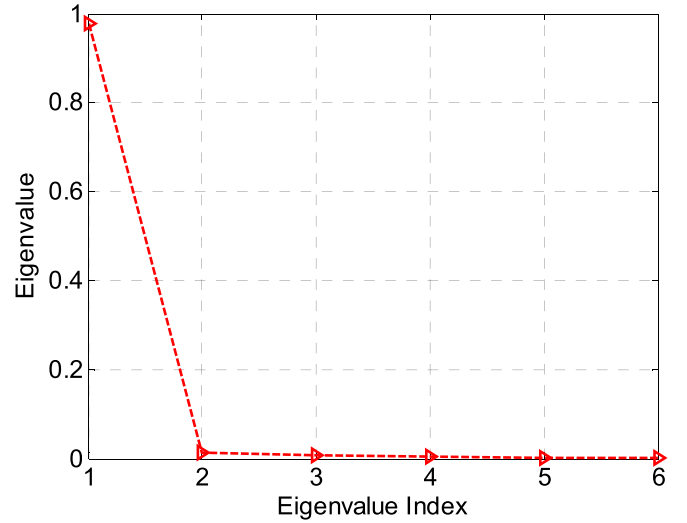
Fig. 3. Pseudocolor images (using $|HH - VV|$, $|HV|$, $|HH + VV|$ as red, green, and blue, respectively). (a) RadarSat-2 data. (b) Region in green window in (a).

Fig. 4. Eigenvalues of the correlation matrix of the sea clutter.

IV. ANALYSIS OF EXTENDED DETECTORS

A. Relation Between Extended Detectors and Traditional Detectors

STD, PTD, and EPTD can be represented by the following united formula

$$\gamma_d = \frac{1}{\sqrt{1 + \text{RedR} \frac{\|[\mathbf{Q}_C]_{\mathbf{P}}\|^2}{\|[\mathbf{Q}_T]_{\mathbf{P}}\|^2}}}. \quad (55)$$

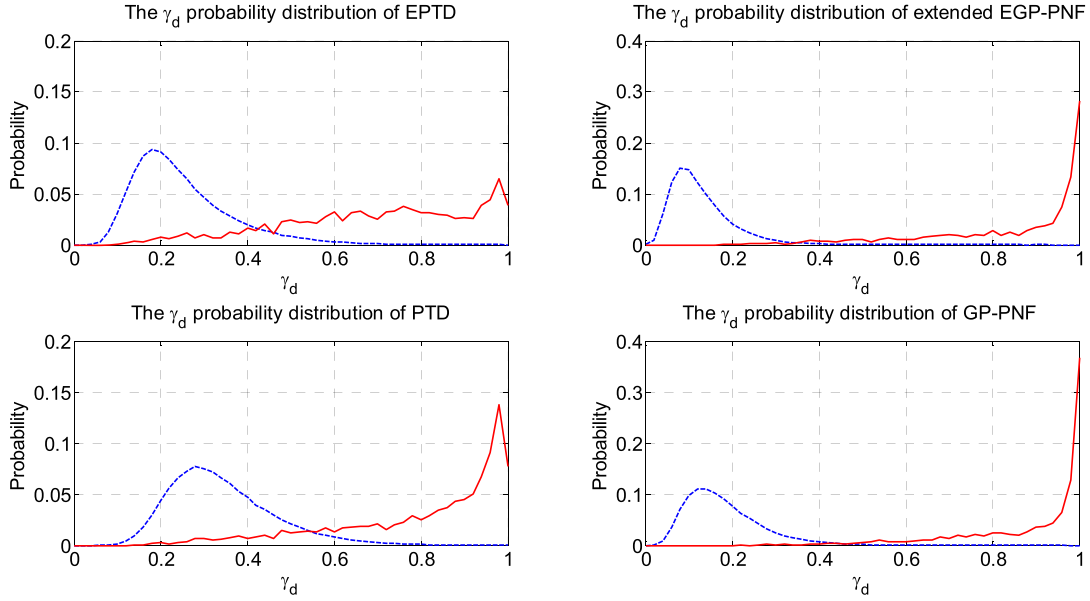


Fig. 5. Probability distributions of γ_d values of different detectors for homogeneous background.

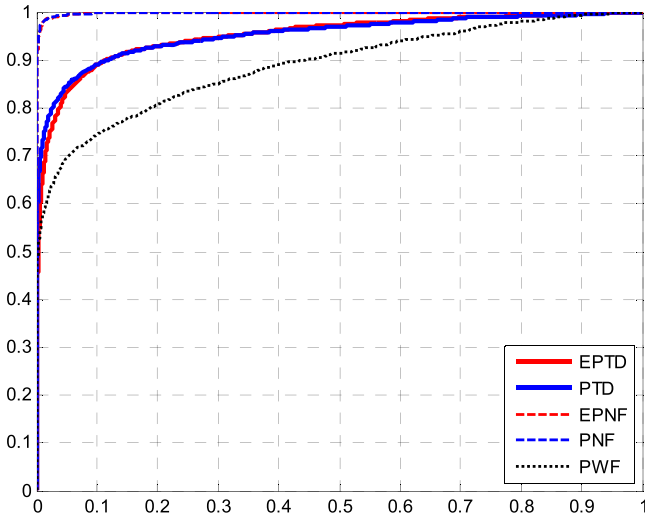


Fig. 6. ROC curves of different detectors for homogeneous background.

In the STD case, $\underline{p}$ is a 3-D complex vector $\underline{k}$; Q_C and Q_T are the orthogonal projection matrices of the clutter subspace S_C and target subspace S_T in $\mathbb{C}^3$, respectively. As mentioned in Section II, STD chooses the dimension of S_T as 1, leading to a 2-D clutter subspace S_C .

In the PTD case, $\underline{p}$ is a 6-D complex vector $\underline{t}$; Q_C and Q_T are the orthogonal projection matrices of the clutter subspace S_C and target subspace S_T in $\mathbb{C}^6$, respectively. PTD also choose the dimension of S_T as 1, leading to a 5-D clutter subspace S_C .

For EPTD, $\underline{p}$ is a 6-D complex vector $\underline{t}$; Q_C and Q_T are the orthogonal projection matrices of the clutter subspace S_C and target subspace S_T in $\mathbb{C}^6$. However, the dimension of S_T can be any integer number between 1 and 5. It is chosen according to the situation. Hence, PTD can be seen as a special case of EPTD.

GP-PNF and EGP-PNF can be represented by the following united formula:

$$\gamma_d = \frac{1}{\sqrt{1 + \frac{\text{RedR}}{\| [Q_T] \underline{t} \|^2}}}. \quad (56)$$

In the GP-PNF case, Q_T is the OPM of the target subspace S_T in $\mathbb{C}^6$. GP-PNF chooses the dimension of S_T as 5, leading to a 1-D clutter subspace S_C .

For EGP-PNF, Q_T is the OPM of the target subspace S_T in $\mathbb{C}^6$. However, the dimension of S_T can be any integer number between 1 and 5. It is chosen according to the situation. Hence, GP-PNF can be seen as a special case of EGP-PNF.

STD, PTD, and EPTD neglect the backscattering intensity and use only the scattering mechanism to distinct target from clutter. Hence STD, PTD, and EPTD are not sensitive to the backscattering intensity. A weak backscattering target can still be detected if only the scattering mechanism of the target is different from the clutter.

GP-PNF and EGP-PNF remove the power that lies in the clutter subspace and use the filtered backscattering intensity to distinct the target from the clutter. Hence, GP-PNF and EGP-PNF are sensitive to the backscattering intensity. GP-PNF and EGP-PNF have high target detection performance for targets that have high backscattering intensity relative to the clutter. However, for the clutter that has high backscattering intensity relative to the target, the clutter may also have high power in the target subspace and cause high false alarm probability.

B. Geometrical Interpretation of the EPTD, PTD, and STD

When RedR in (55) is fixed, γ_d is completely determined by $\| [Q_C] \underline{p} \|^2 / \| [Q_T] \underline{p} \|^2$, which can be rewritten as

$$\frac{\| [Q_C] \underline{\hat{p}} \|^2}{\| [Q_T] \underline{\hat{p}} \|^2} \quad (57)$$

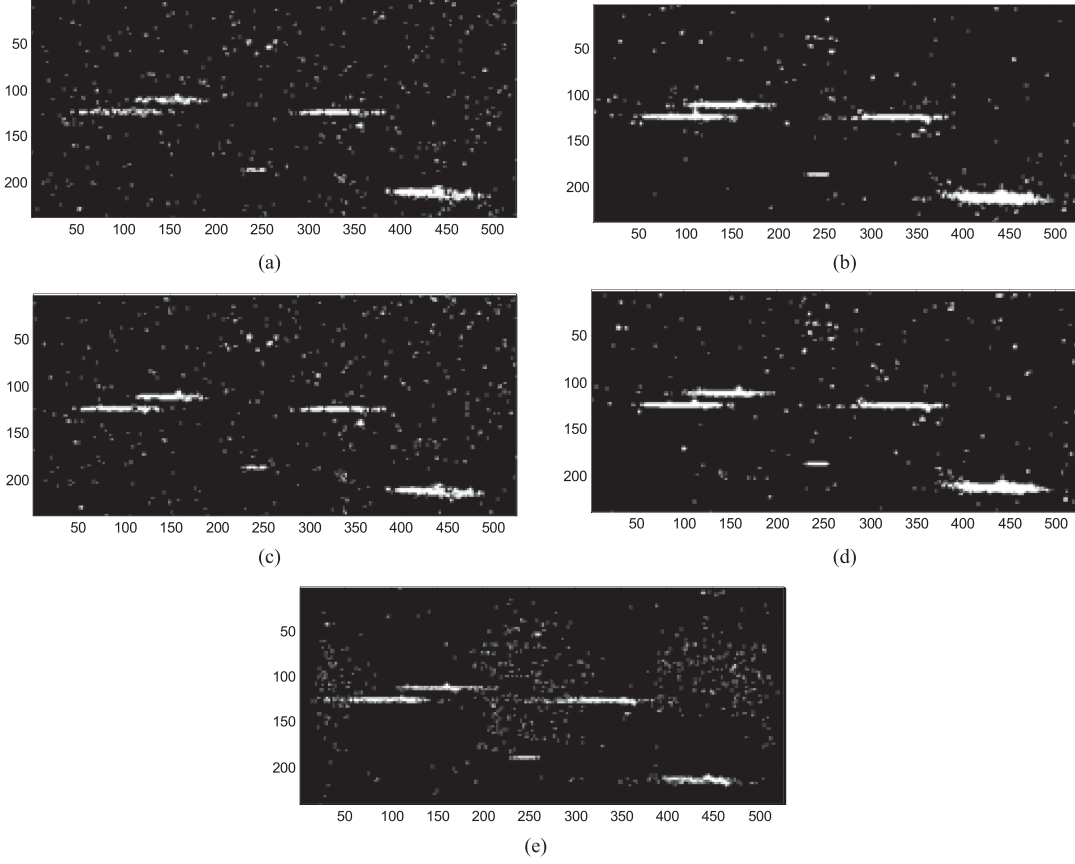


Fig. 7. Detection results of different detectors for homogeneous background. (a) EPTD. (b) EGP-PNF. (c) PTD. (d) PNF. (e) PWF.

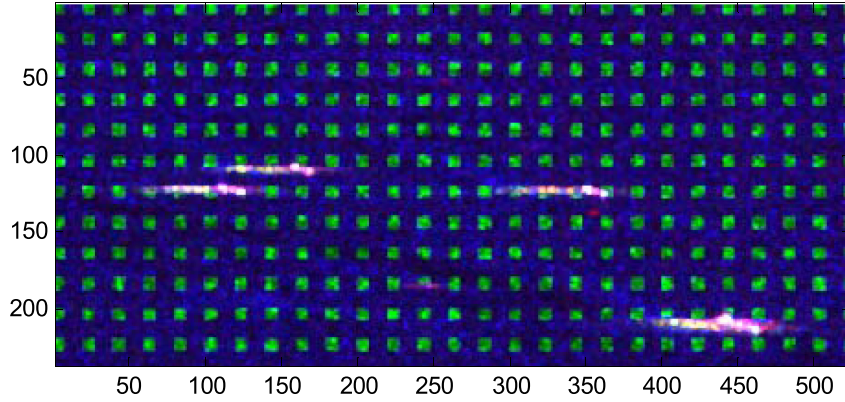


Fig. 8. Pseudocolor image (using $|HH - VV|$, $|HV|$, $|HH + VV|$ as red, green, and blue, respectively) of the region in green window in Fig. 3(a) after embedded with extra *oriented dihedral* patches.

where $\hat{p}$ is $\underline{p}/\|\underline{p}\|^2$. Let θ be the principal angle between subspace $\text{span}(\hat{p})$ and the clutter subspace S_C . Then

$$\|[Q_C]\hat{p}\|^2 = 1 - d(S_C, \text{span}(\hat{p}))^2 \quad (58)$$

where $d(S_C, \text{span}(\hat{p}))$ stands for the distance between the subspace $\text{span}(\hat{p})$ and the clutter subspace S_C ; $d(S_C, \text{span}(\hat{p})) = \sin(\theta)$. For the definitions of the principal angle and the distance between subspaces of different dimensions, one can refer to [33].

Since

$$\|[Q_C]\hat{p}\|^2 + \|[Q_T]\hat{p}\|^2 = 1. \quad (59)$$

Then, (57) can be rewritten as follows:

$$\frac{\|[Q_C]\hat{p}\|^2}{\|[Q_T]\hat{p}\|^2} = \frac{\|[Q_C]\hat{p}\|^2}{1 - \|[Q_C]\hat{p}\|^2} = \frac{1 - d(S_C, \text{span}(\hat{p}))^2}{d(S_C, \text{span}(\hat{p}))^2}. \quad (60)$$

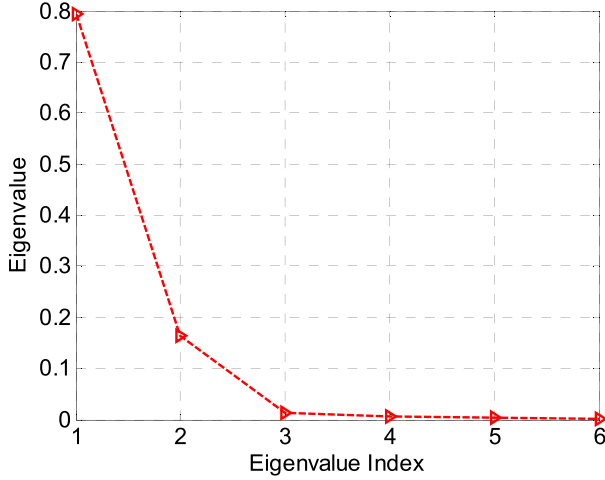


Fig. 9. Eigenvalues of the correlation matrix of the mixed clutter simulated with oriented dihedral.

Apparently, (57) can also be represented in terms of $d(S_T, \text{span}(\hat{p}))$ as follows:

$$\frac{\| [Q_C] \hat{p} \|^2}{\| [Q_T] \hat{p} \|^2} = \frac{1 - \| [Q_T] \hat{p} \|^2}{\| [Q_T] \hat{p} \|^2} = \frac{d(S_T, \text{span}(\hat{p}))^2}{1 - d(S_T, \text{span}(\hat{p}))^2}. \quad (61)$$

It can be seen from (41) that γ_d is large when $\| [Q_C] \hat{p} \|^2 / \| [Q_T] \hat{p} \|^2$ is small. Also, from (60) and (61), it can be obtained that γ_d is large when $d(S_C, \text{span}(\hat{p}))$ is large, in other words, when $d(S_T, \text{span}(\hat{p}))$ is small. That is, EPTD, PTD, and STD are used to detect the partial scattering vectors that are far from the clutter subspace, in other words, close to the target subspace.

V. VALIDATION

A. Introduction of Datasets

In order to provide a validation of EPTD and EGP-PNF, a C-band RADARSAT-2 Fine-Quad-Mode full PolSAR dataset is employed. The resolutions are approximately 8m and 12m in the azimuth and range directions, respectively. The data were acquired on August 4, 2010.

PTD, GP-PNF, and PWF are used for comparison with the proposed EPTD and EGP-PNF.

It should be noted that PTD is used in a different way with how Marino used in [11], [18] and [20]. Marino use a partial scattering vector to represent the targets to be detected and exploit PTD to detect the targets. In this paper, we use a partial scattering vector $\hat{t}_{\text{clutter}}$ that is estimated automatically to represent the clutter and then detect the target different from the partial scattering vector. That is, we use the following formula in this experiments:

$$\gamma_d = \frac{1}{\sqrt{1 + \text{RedR} \frac{|\hat{t}^{*T} \hat{t}_{\text{clutter}}|^2}{\hat{t}^{*T} \hat{t} - |\hat{t}^{*T} \hat{t}_{\text{clutter}}|^2}}}. \quad (62)$$

PWF is used as a comparison here because PWF has the best performance among algorithms without a prior information expect GP-PNF [13]. In addition, PTD, GP-PNF, PWF, EPTD, and EGP-PNF all need to estimate the clutter automatically in the experiments. Hence, it is fair to compare the target detection performance of the five detectors.

B. Experiments for Homogeneous Background

The RGB pseudocolor image of the data used in this experiment is shown in Fig. 3(a). The region in the green window includes five ships and sea clutter. And it is used to test the detectors. Fig. 3(b) shows the pseudocolor image of the region in the green window.

First, the heterogeneity of the sea clutter is analyzed. The correlation matrix of the feature vectors of the background of the chip in green window is calculated as in (50). The eigenvalues of the correlation matrix sorted from the largest to the smallest are shown in Fig. 4. It can be seen that the second largest eigenvalue is far less than the largest one, which proves the homogeneity of the sea clutter. Hence, it is suitable for the background to be described with a vector.

To compare the performance of the detectors, EPTD, EGP-PNF, PTD, GP-PNF, and PWF are used to detect the targets in the test chip. For EPTD, the dimension of clutter subspace is chosen as 2 and the RedR as 0.2. For EGP-PNF, the dimension of clutter subspace is chosen as 2 and the RedR as $1E-4$. For PTD, the RedR is chosen as 0.2, and for GP-PNF, the RedR is chosen as $1E-4$. The sliding window method is used to estimate the clutter in all the five detectors. The target window has 4 rows and 20 columns; the guard ring has 15 rows and 101 columns; and the boundary ring has 51 rows and 151 columns. The cells in the boundary ring are used to train EPTD, EGP-PNF, PTD, GP-PNF, and PWF.

The distributions of sea clutter γ_d values and target γ_d values of EPTD, EGP-PNF, PTD, and GP-PNF are shown in Fig. 5. From the figure, it can be seen that the clutter γ_d values obtained by EPTD are relatively smaller compared with the traditional PTD, and the clutter γ_d values obtained by EGP-PNF are relatively smaller compared with the traditional GP-PNF. That means the extended detectors have a better clutter suppression performance than traditional ones. However, the target γ_d values obtained by EPTD and EGP-PNF are also relatively smaller compared with traditional PTD and GP-PNF. This is because when extending the clutter subspace from 1-D to 2-D, we improve our capability to reject it; however, we also inevitably remove part of the target power that projects over the clutter subspace. It can also be observed that EGP-PNF and PNF have larger target γ_d values relative to EPTD and PTD. This is because EGP-PNF and PNF use backscattering intensity as the feature, and the backscattering intensity of the ships is large compared with the clutter.

Fig. 6 shows the receiver operating characteristic (ROC) curves of the five detectors. And the detection results of the five detectors with a false alarm rate of 0.01 are shown in Fig. 7. It is observed that EPTD and PTD almost have the same detection performance; EGP-PNF and PNF almost the same detection performance. This is because the sea clutter is homogeneous,

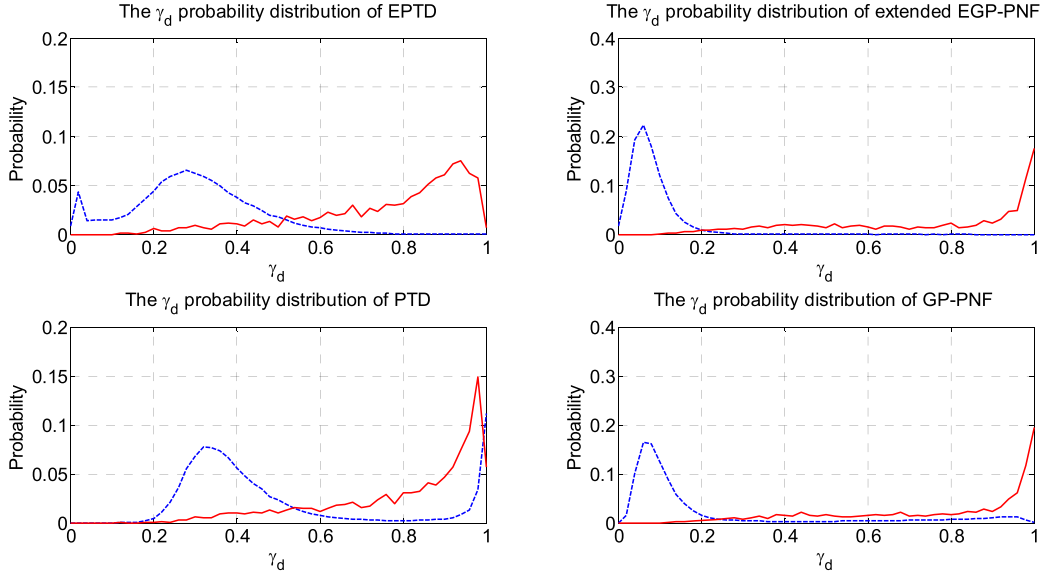


Fig. 10. Probability distributions of γ_d values of different detectors for heterogeneous background simulated with oriented dihedral.

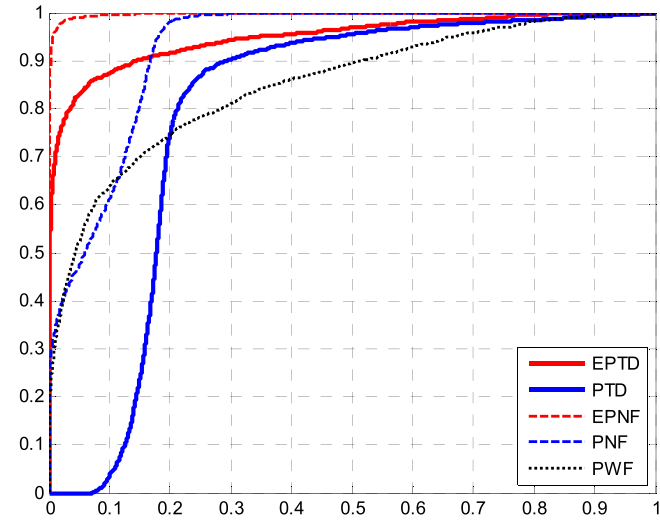


Fig. 11. ROC curves of different detectors for heterogeneous background simulated with oriented dihedral.

and in such case, we are reducing the probability of false alarms at the expenses of reducing the probability of detection. The overall performance is therefore expected to stay the same. On the other hand, EGP-PNF and GP-PNF have better detection performance than EPTD and PTD. The reason is, as mentioned previously, EGP-PNF and GP-PNF exploit the scattering mechanism and backscattering intensity to distinguish target from the clutter. And the backscattering intensity is an important feature in this scene since the backscattering intensity of the ships is far higher than that of the sea clutter. It is also observed that the ROC curve of PWF is not so good as the other four detectors.

C. Experiments for Simulated Heterogeneously Patched Background

1) *Simulation With Oriented Dihedral:* In order to test EGP-PNF and GP-PNF in heterogeneously patched background, we

simulate a 7×7 patch and embed it into the sea background of the chip shown in Fig. 3(b). The pixels of the patch have Sinclair matrix in the following form:

$$S = A \begin{bmatrix} 0 & 1/\sqrt{2} \\ 1/\sqrt{2} & 0 \end{bmatrix} \quad (63)$$

where A is the backscattering intensity. A^2 equals the span of the sea pixel which the simulated pixel replaces. The pseudocolor image of the finally obtained chip is shown in Fig. 8.

The mixed background is heterogeneous, because there exists more than one scattering mechanism in the background. This can be easily seen from the eigenvalues of the correlation matrix of the features vectors of mixed background. The eigenvalues of the correlation matrix sorted from the largest to the smallest are shown in Fig. 9. The second largest eigenvalue is comparable to the largest one, which proves that there are two scattering mechanisms in the mixed background and the mixed background is heterogeneous.

EPTD, EGP-PNF, PTD, GP-PNF, and PWF are used to detect the targets in mixed background. The configuration of the detectors and the sliding window is the same as the experiment in Section V-B.

The probability distributions of sea clutter γ_d values and target γ_d values of EPTD, EGP-PNF, PTD, and GP-PNF are shown in Fig. 10. From the figure, it can be seen that EPTD has better clutter suppression performance than PTD, whereas EGP-PNF has better clutter suppression performance than GP-PNF. This is because the EPTD and EGP-PNF use a 2-D subspace to represent the clutter.

Fig. 11 shows the ROC curves of the five detectors. The detection results of the five detectors with a false alarm rate of 0.01 are shown in Fig. 12. These results demonstrate that EPTD and EGP-PNF outperform PTD, GP-PNF, and PWF for target detection in heterogeneously patched background.

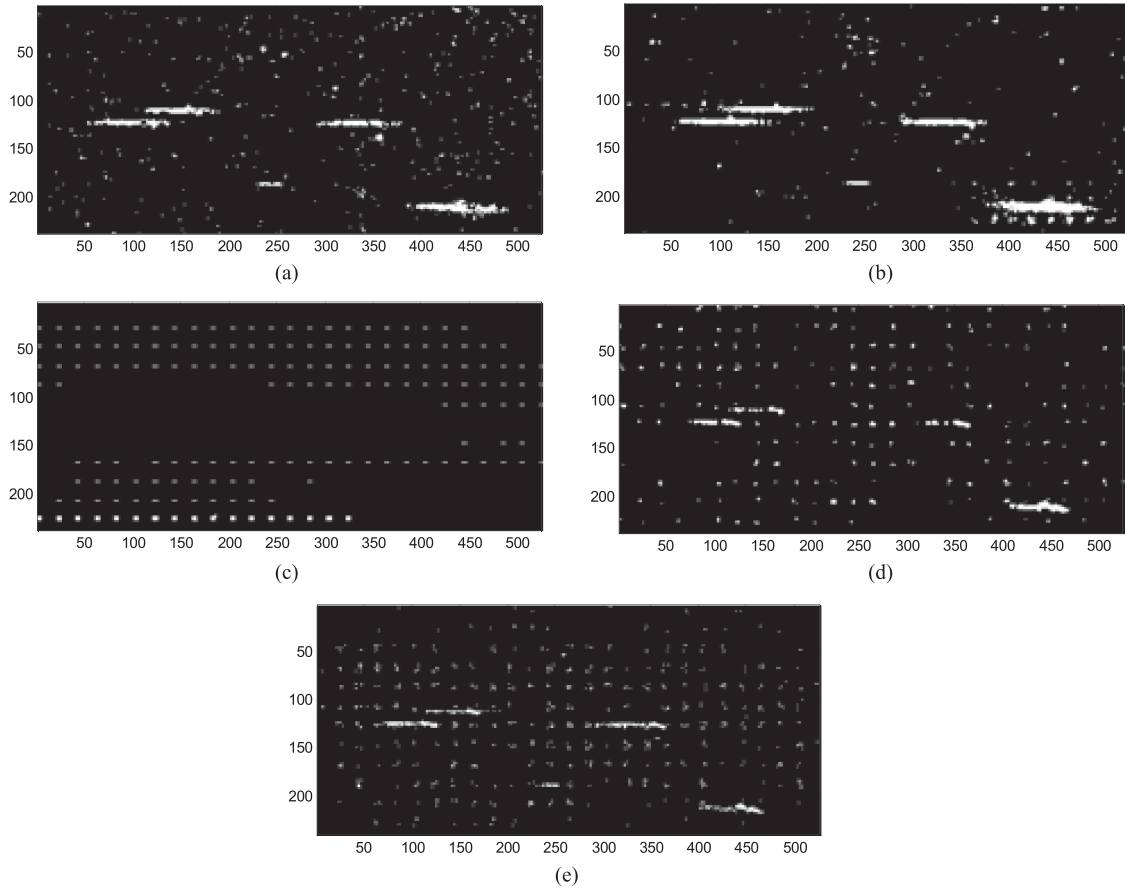


Fig. 12. Detection results of different detectors with a false alarm rate of 0.01 for heterogeneous background simulated with oriented dihedral. (a) EPTD. (b) EGP-PNF. (c) PTD. (d) PNF. (e) PWF.

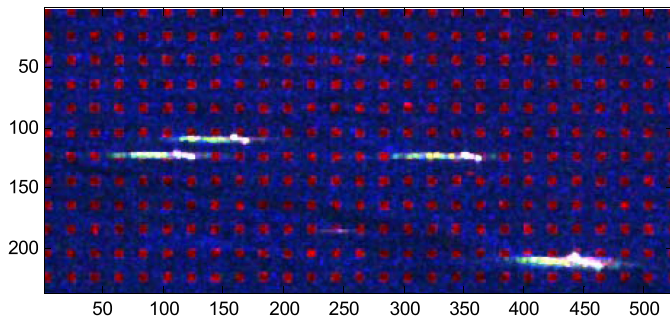


Fig. 13. Pseudocolor image (using $|HH - VV|$, $|HV|$, $|HH + VV|$ as red, green, and blue, respectively) of the region in green window in Fig. 3(a) after embedded with extra horizontal dihedral patches.

It is also observed that EGP-PNF outperforms EPTD. The reason is that EGP-PNF exploits the scattering mechanism and backscattering intensity to distinguish the target from the clutter. And the backscattering intensity is a good feature in such scene, since the backscattering intensity of the ships is far higher than the clutter.

2) *Simulation With Horizontal Dihedral*: In order to further test EGP-PNF and GP-PNF in heterogeneously patched background, we simulate a 7×7 patch with a different scattering mechanism and embed it into the sea background of the chip

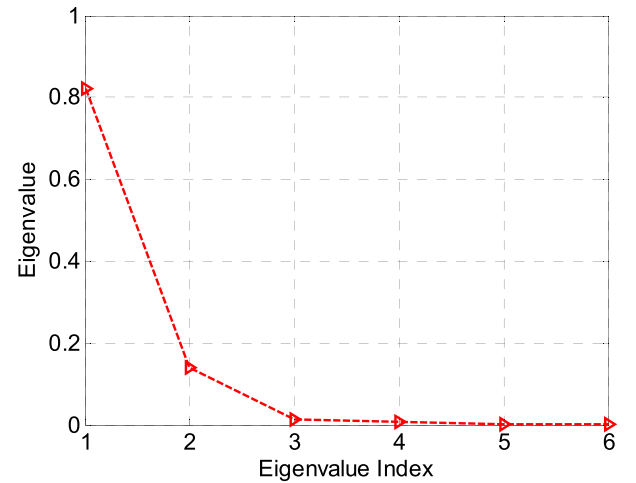


Fig. 14. Eigenvalues of the correlation matrix of the mixed clutter simulated with horizontal dihedral.

shown in Fig. 3(b). The pixels of the patch have Sinclair matrix in the following form;

$$S = A \begin{bmatrix} 1/\sqrt{2} & 0 \\ 0 & -1/\sqrt{2} \end{bmatrix} \quad (64)$$

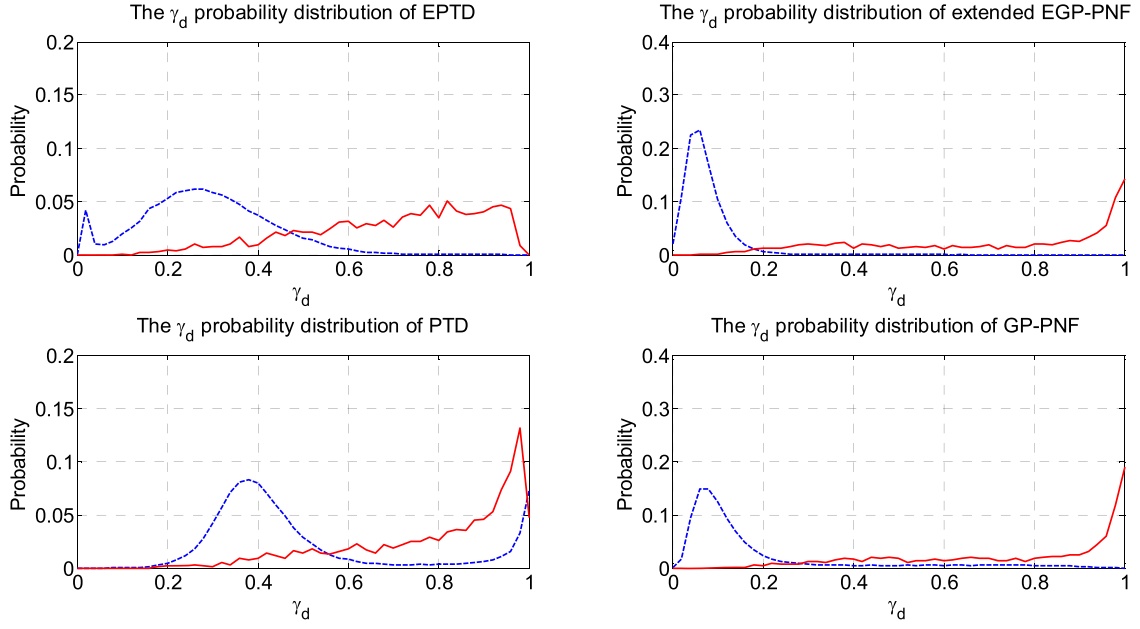


Fig. 15. Probability distribution of γ_d values of different detectors for heterogeneous background simulated with horizontal dihedral.

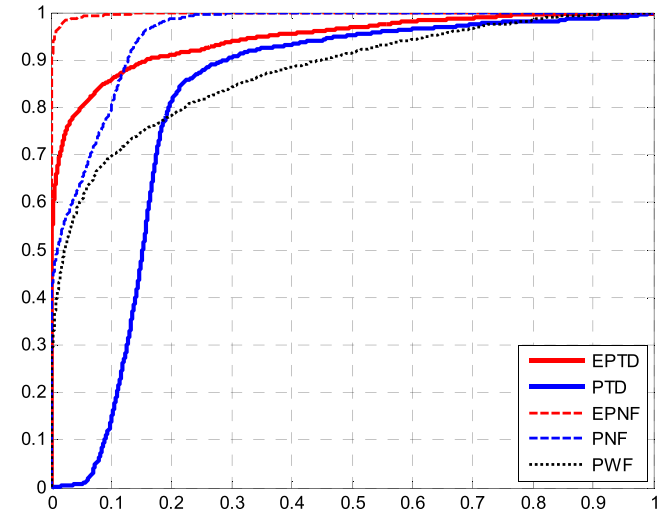


Fig. 16. ROC curves of different detectors for heterogeneous background simulated with horizontal dihedral.

where A is the backscattering intensity. A^2 equals the span of the sea pixel which the simulated pixel replaces. The pseudocolor image of the finally obtained chip is shown in Fig. 13.

The experiment configurations are the same as the experiment in Section V-C-1. The eigenvalues of the correlation matrix sorted from the largest to the smallest are shown in Fig. 14. The second largest eigenvalue is comparable to the largest one, which proves that there are two scattering mechanisms in the mixed background and the mixed background is heterogeneous. Fig. 15 shows the ROC curves of the five detectors. The detection results of the five detectors with a false alarm rate of 0.01 are shown in Fig. 16. It is observed that the experimental results are very close to the results in Section V-C-1. These results demonstrate

that EPTD and EGP-PNF outperform PTD, GP-PNF, and PWF for target detection in heterogeneously patched background.

By comparing Fig. 17(a) with Fig. 12(a), it is observed that Fig. 17(a) detects less target pixels than Fig. 12(a). The reason is that some pixels of the ships have scattering mechanisms close to horizontal dihedral, which results in the distance between the partial scattering vectors of the pixels and the clutter subspace very close. Therefore, these pixels are hard to be detected by EPTD.

D. Experiments for Measured Heterogeneously Patched Background

In order to test extend GP-PNF and GP-PNF in measured heterogeneously patched background, we extract a 7×7 patch from the region in the red window in Fig. 3(a) and embed it into the sea background of the chip shown in Fig. 3(b). The pseudocolor image of the finally obtained chip is shown in Fig. 18.

The mixed background is heterogeneous, because there exists more than one scattering mechanism in the background. This can be easily seen from the eigenvalues of the correlation matrix of the features vectors of mixed background. The eigenvalues of the correlation matrix sorted from the largest to the smallest are shown in Fig. 19. The second largest eigenvalue is comparable to the largest one, which proves that there are two main scattering mechanisms in the mixed background and the mixed background is heterogeneous. Hence, it is not suitable for the clutter in the mixed background to be described with a vector.

EPTD, EGP-PNF, PTD, GP-PNF, and PWF are used to detect the targets in mixed background. The configurations of the detectors and the sliding window are the same as the experiment in Section V-B.

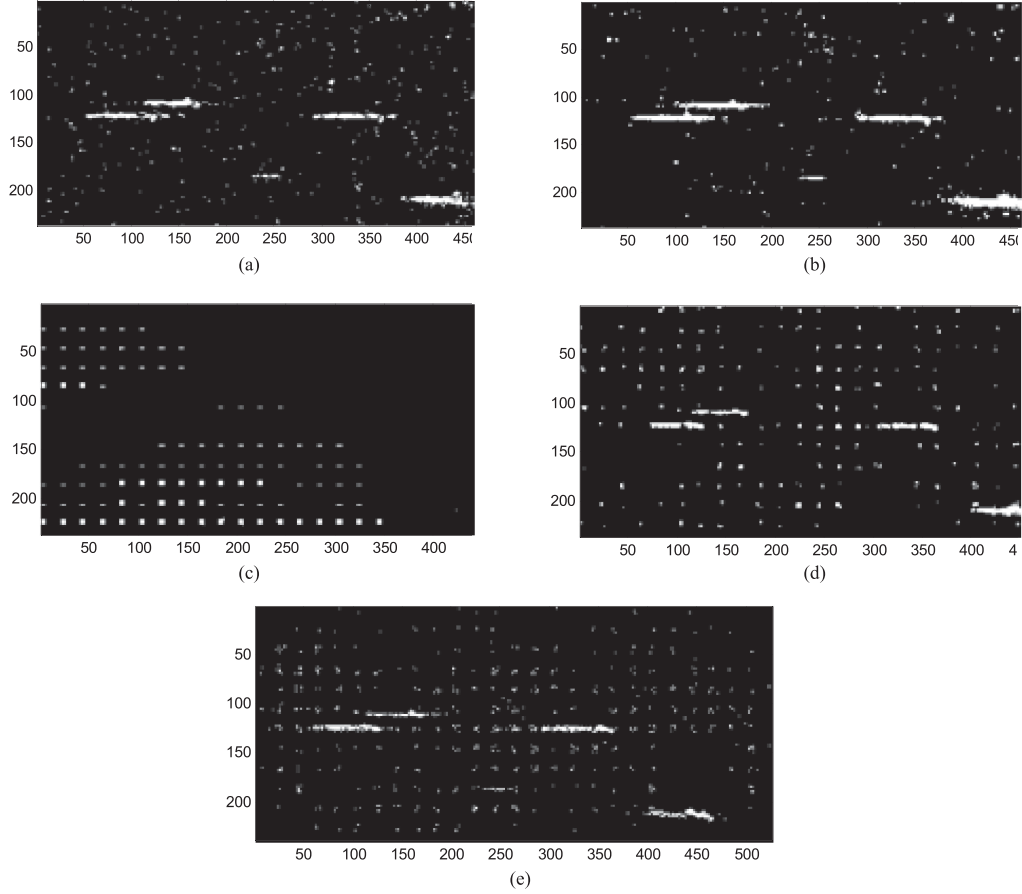


Fig. 17. Detection results of different detectors with a false alarm rate of 0.01 for heterogeneous background simulated with horizontal dihedral. (a) EPTD. (b) EGP-PNF. (c) PTD. (d) PNF. (e) PWF.

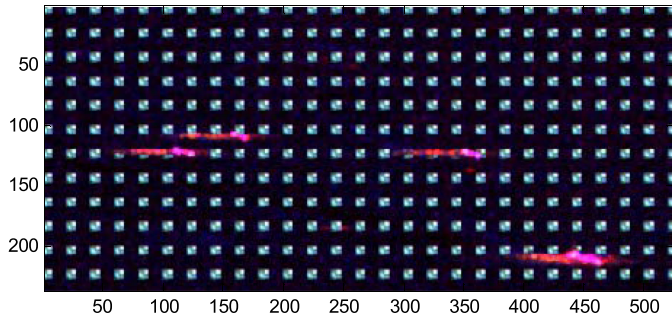


Fig. 18. Pseudocolor image (using $|HH - VV|$, $|HV|$, $|HH + VV|$ as red, green, and blue, respectively) of the region in green window in Fig. 3(a) after embedded with extra measured patches.

The probability distributions of sea clutter γ_d values and target γ_d values of EPTD, EGP-PNF, PTD, and GP-PNF are shown in Fig. 20. Fig. 21 shows the ROC curves of the five detectors. The detection results of the three detectors with a false alarm rate of 0.01 are shown in Fig. 22. These results demonstrate that EPTD outperforms EGP-PNF, PTD, GP-PNF, and PWF for target detection in such background.

From Fig. 20, it is observed that there exist lots of clutter pixels that have large γ_d values for EGP-PNF, PTD, and GP-PNF. Now we discuss the reasons. Suppose the OPM of the

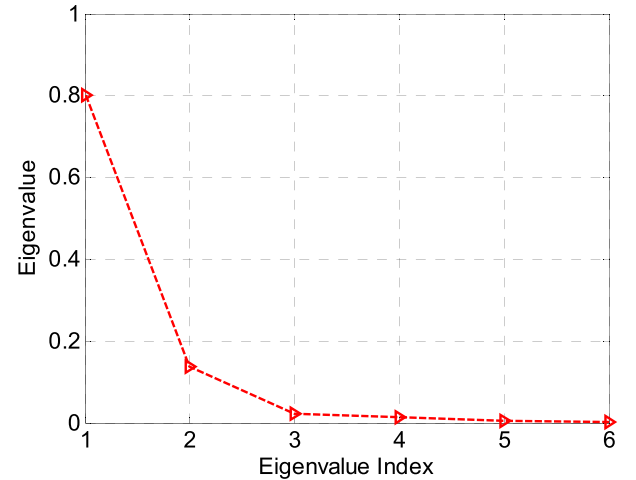


Fig. 19. Eigenvalues of the correlation matrix of the mixed clutter simulated with measured patches.

clutter subspace is Q_C . For GP-PNF and EGP-PNF, the power that a clutter partial scattering vector $\underline{t}_C$ leaks into the target subspace is

$$P_{\text{leak1}} = \|([I] - [Q_C]) \underline{t}_C\|_2^2 \quad (65)$$

where $[I]$ is identity matrix. For PTD and EPTD, since they do not care about the back scattering intensity, the power

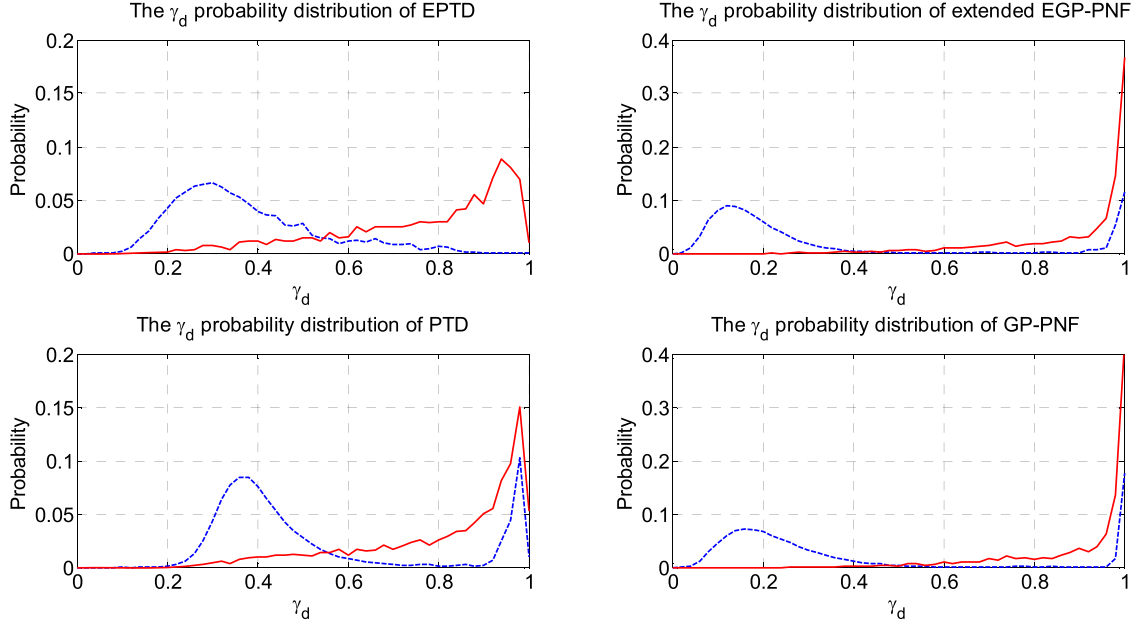


Fig. 20. Probability distributions of γ_d values of different detectors for heterogeneous background simulated with measured patches.

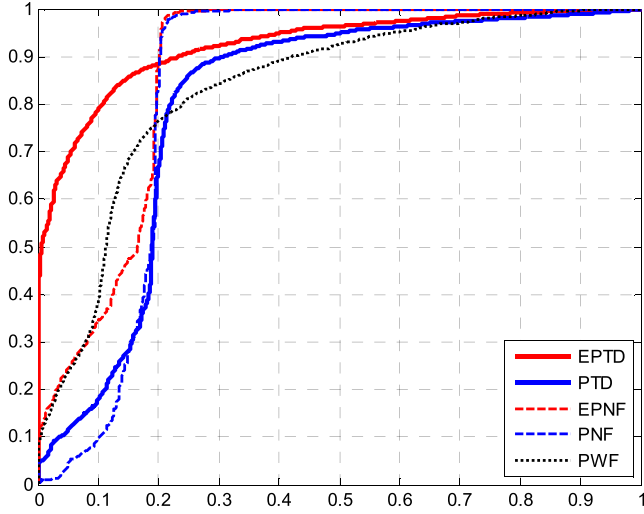


Fig. 21. ROC curves of different detectors for heterogeneous background simulated with measured patches.

that a clutter partial scattering vector $\hat{t}_C$ leaks into the target subspace is

$$P_{\text{leak2}} = \|([I] - [Q_C])\hat{t}_C\|_2^2 \quad (66)$$

where $\hat{t}_C = t_C / \|t_C\|_2^2$. In the ideal case, the clutter vectors completely lie in the clutter subspace (i.e., the distance between the clutter vector and the clutter subspace is 0), which makes $P_{\text{leak1}} = P_{\text{leak2}} = 0$. However, in practice, the distance between the clutter vector and the clutter subspace is usually not 0, which makes part power of the clutter vector leaking into the target subspace. By observing (65), it is found that for GP-PNF and EGP-PNF, the further the clutter vector from the clutter subspace is and the higher the backscattering intensity of the clutter vector

has, the more the power of the clutter vector leaks into the target subspace. By observing (66), it is found that for PTD and EPTD, the leaking power only depends on the distance between the clutter vector and the clutter subspace. The further the clutter vector is from the clutter subspace, the more the power of the clutter vector leaks into the target subspace. To clarify, the distance between a vector and a subspace means the distance of the subspace spanned by the vector and the subspace.

For PTD, there exist lots of clutter pixels that have large γ_d values because the background is heterogeneous and it is not suitable to represent the clutter with a partial scattering vector (i.e., 1-D subspace). When we use a 1-D subspace to represent the background, some clutter vectors are far from the clutter subspace, which results in the high power of these clutter vectors leaking into the target subspace. And the leaking power makes the clutter vectors have high γ_d values.

For EPTD, there are few clutter pixels have γ_d values. This is due to that when we use a 2-D subspace to represent the background, most of the clutter vectors are close to the clutter subspace. Compared with PTD, the clutter vectors are leaking less power to the target subspace.

For EGP-PNF, although the clutter vectors are close to the clutter subspace, the backscattering intensity of some clutter vectors is high. Hence, these clutter vectors have high power leaking into the target subspace, which makes the clutter vectors have high γ_d value. Apparently the backscattering intensity is not a good feature in such scene.

For PNF, there exist lots of clutter pixels that have large γ_d values because of two reasons: It uses a partial scattering vector to represent the clutter and the background has some clutter pixels with high backscattering intensity.

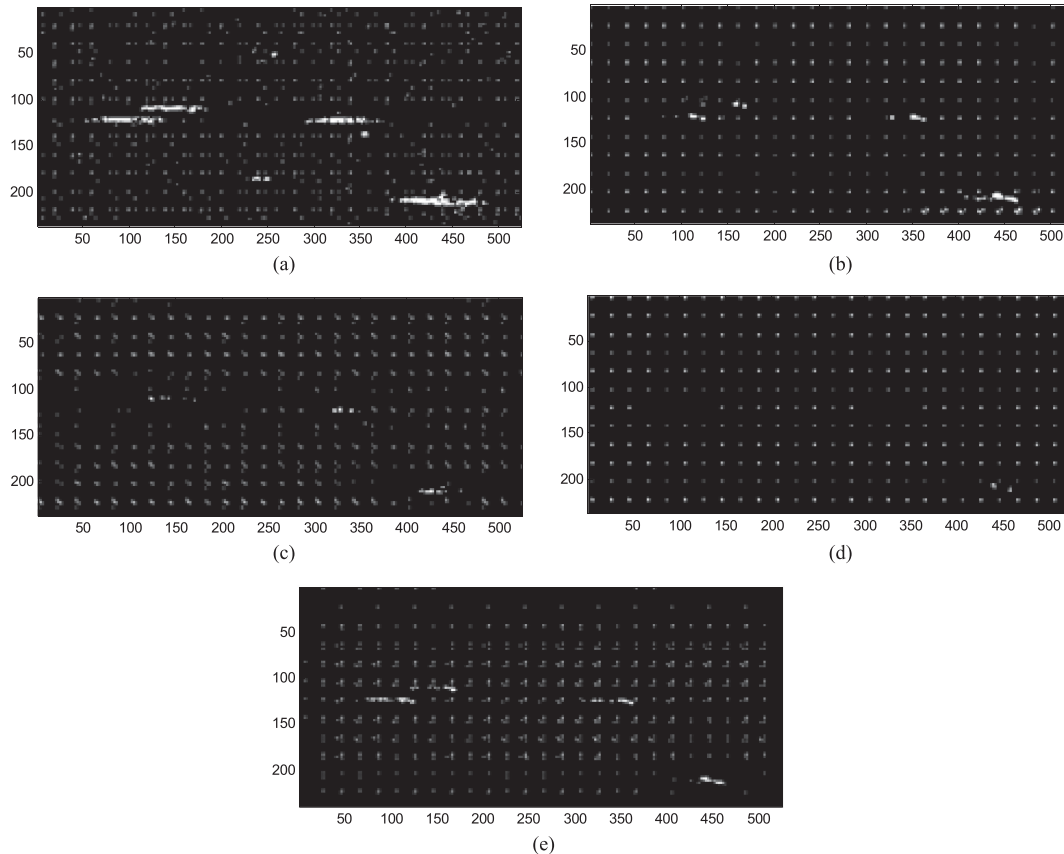


Fig. 22. Detection results of different detectors with a false alarm rate of 0.01 for heterogeneous background simulated with measured patches. (a) EPTD. (b) EGP-PNF. (c) PTD. (d) PNF. (e) PWF.

VI. CONCLUSION

This paper proposes EPTD and EGP-PNF, which are generalization of PTD and GP-PNF, respectively. EPTD and EGP-PNF describe the target as a subspace in $\mathbb{C}^6$, which allows them to describe heterogeneous clutter. By this way, EPTD and EGP-PNF can detect targets in the heterogeneous background. The dimension of the clutter subspace is usually set to be less than 3, since there only exists three orthogonal partial scattering vectors that are physically feasible.

Experimental results demonstrate that in homogeneous background, EPTD has the same performance with PTD, and EGP-PNF has the same performance with GP-PNF; in heterogeneous background, EPTD and EGP-PNF outperform PTD, GP-PNF, and PWF. On the other hand, compared with EPTD, EGP-PNF has better performance in detecting targets that have large backscattering intensity relative to the clutter and have worse performance in detecting targets that do not have large backscattering intensity relative to the clutter.

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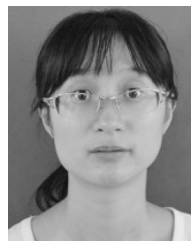
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